Notes

Introduction and Revision

Welcome and Revision

Hello everyone! I hope you all are doing well and have revised the concepts covered yesterday. We will continue our discussion on motion in one dimension.

Recap of Previous Session

Yesterday, we derived three fundamental equations of motion. Today, we will derive the formula for displacement in the nth second. Following that, we will solve numerous problems and I will also share some assignments with you all.

Displacement in the nth Second: Concept and Derivation

Understanding Displacement in the nth Second

Displacement in the nth second refers to the displacement of an object during a specific one-second interval. For example:

- **Displacement in the 6th second (S_6)**: This means the displacement from t = 5 seconds to t = 6 seconds.
- **Displacement in the 8th second (S_8)**: This means the displacement from t = 7 seconds to t = 8 seconds.
- **Displacement in the 4th second (S_4)**: This means the displacement from t = 3 seconds to t = 4 seconds.
- **Displacement in the 10th second (S_{10})**: This means the displacement from t = 9 seconds to t = 10 seconds.

In general, the displacement in the nth second (S_n) is the displacement from $t = n - 1$ seconds to $t = n$ seconds.

Derivation of the Formula for Displacement in the nth Second

To find the displacement in the nth second, we can use the following approach:

S_n = Displacement till n seconds − Displacement till (n-1) seconds

We know the general formula for displacement under constant acceleration is:

$S = ut + \frac{1}{2}at^2$

Where:

- t = time

1. **Displacement till n seconds (S_n)**: Substitute $t = n$ into the displacement formula: $S_n = un + \frac{1}{2}an^2$
2. **Displacement till (n-1) seconds (S_{n-1})**: Substitute $t = n - 1$ into the displacement formula: $S_{n-1} = u(n - 1) + \frac{1}{2}a(n - 1)^2$

Now, subtract S_{n-1} from S_n:

$S_{nth} = (un + \frac{1}{2}an^2) - [u(n - 1) + \frac{1}{2}a(n - 1)^2]$

Expand the terms:

$S_{nth} = un + \frac{1}{2}an^2 - [un - u + \frac{1}{2}a(n^2 - 2n + 1)]$

$S_{nth} = un + \frac{1}{2}an^2 - [un - u + \frac{1}{2}an^2 - an + \frac{1}{2}a]$

Distribute the negative sign:

$S_{nth} = un + \frac{1}{2}an^2 - un + u - \frac{1}{2}an^2 + an - \frac{1}{2}a$

Cancel out the un and $\frac{1}{2}an^2$ terms:

$S_{nth} = u + an - \frac{1}{2}a$

Rearrange the terms to factor out $a/2$:

$S_{nth} = u + a(n - \frac{1}{2})$

$S_{nth} = u + a(\frac{2n-1}{2})$

$S_{nth} = u + \frac{a}{2}(2n - 1)$

This is the formula for the displacement in the nth second.

Application of the Formula

To find the displacement in any specific nth second (e.g., the 6th second), simply substitute the value of n into the formula, along with the initial velocity (u) and acceleration (a).

- u: Initial velocity of the object.
- a: Constant acceleration of the object.
- n: The specific second for which the displacement is being calculated (e.g., 6 for the 6th second).

For example, to find the displacement in the 6th second, you would put $n = 6$ in the formula.

Kinematic Equations

We will be using the three fundamental kinematic equations to solve problems related to motion in a straight line:

1. $v = u + at$
2. $s = ut + \frac{1}{2}at^2$
3. $v^2 = u^2 + 2as$

Understanding Retardation

Retardation refers to a situation where the speed of an object decreases. It does not necessarily mean negative acceleration. The sign of acceleration (positive or negative) depends on the chosen coordinate system and the direction of motion.

- If we define the right direction as positive, and an object is moving to the right, then an acceleration to the left would cause retardation.
- The plus (+) and minus (-) signs in physics primarily indicate direction.

Problem: Car Braking - Finding Time to Stop and Stopping Distance

Problem Statement

A car is initially moving at a speed of 40 m/s. The driver applies the brakes, causing a retardation of 4 m/s^2. We need to find:

1. The time it takes for the car to stop.
2. The stopping distance (the distance covered by the car until it stops).

Given Information

- Initial velocity (u) = 40 m/s
- Final velocity (v) = 0 m/s (since the car stops)
- Retardation = 4 m/s^2

Setting up the Coordinate System

Let's assume the direction of the car's initial motion (to the right) is the positive direction. Therefore:

- Initial velocity (u) = $+40$ m/s
- Since there is retardation, the acceleration is in the opposite direction to the velocity. So, acceleration (a) = -4 m/s^2.

- Substitute the known values: $0 = 40 + (-4)t$
- Rearrange the equation to solve for t: $4t = 40$ $t = \frac{40}{4}$ $t = 10$ s

Therefore, the car will stop in 10 seconds.

Part 2: Finding the Stopping Distance (s)

We can use two methods to find the stopping distance.

Method 1: Using $s = ut + \frac{1}{2}at^2$

- Substitute the known values: $s = (40)(10) + \frac{1}{2}(-4)(10)^2$ $s = 400 + \frac{1}{2}(-4)(100)$ $s = 400 - 2(100)$ $s = 400 - 200$ $s = 200$ m

Method 2: Using $v^2 = u^2 + 2as$

- Substitute the known values: $0^2 = (40)^2 + 2(-4)s$ $0 = 1600 - 8s$
- Rearrange the equation to solve for s: $8s = 1600$ $s = \frac{1600}{8}$ $s = 200$ m

Both methods yield the same result. The stopping distance is 200 meters.

Understanding De-acceleration and Retardation

De-acceleration and Retardation

- **De-acceleration** refers to the decrease in speed of an object.
- **Retardation** is synonymous with de-acceleration; both terms mean the same thing.
- There is no difference in meaning between de-acceleration and retardation; both indicate that the speed of an object is decreasing.

Problem: Fly Oscillating Between Bus and Wall

Problem Description

A fly oscillates between a bus and a wall at a constant speed of 200 m/s. The bus starts with an initial speed of 10 m/s and an acceleration of 2 m/s^2. The wall is located 100 m ahead of the bus. We need to find the total distance traveled by the fly before it dies (i.e., before the bus hits the wall).

Scenario Breakdown

- The bus moves towards the wall, constantly reducing the distance between itself and the wall.
- The fly continuously travels back and forth between the bus and the wall.

Notes

To find the total distance traveled by the fly, we first need to determine the total time until the bus reaches the wall. During this entire time, the fly is in motion.

We can use the second equation of motion for the bus: $S = UT + \frac{1}{2}AT^2$ Where:

- S = 100 m (distance to the wall)
- U = 10 m/s (initial speed of the bus)
- A = 2 m/s^2 (acceleration of the bus)
- T is the time taken for the bus to reach the wall.

Substituting the values into the equation: $100 = (10)T + \frac{1}{2}(2)T^2$ $100 = 10T + T^2$ Rearranging the terms to form a quadratic equation: $T^2 + 10T - 100 = 0$

Solving the Quadratic Equation for T

Using the quadratic formula $T = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$:

- a = 1
- b = 10
- c = −100

$T = \frac{-10 \pm \sqrt{(10)^2-4(1)(-100)}}{2(1)}$ $T = \frac{-10 \pm \sqrt{100+400}}{2}$ $T = \frac{-10 \pm \sqrt{500}}{2}$ $T = \frac{-10 \pm 10\sqrt{5}}{2}$

Since time cannot be negative, we reject the negative root: $T = \frac{-10+10\sqrt{5}}{2}$ $T = 5\sqrt{5} - 5$ $T = 5(\sqrt{5} - 1)$ seconds This is the total time until the bus hits the wall.

Calculating Total Distance Traveled by the Fly

The fly moves at a constant speed of 200 m/s throughout the entire duration until the bus hits the wall. Since the speed is constant, the total distance traveled by the fly can be calculated using the formula:

Distance = Speed × Time

- Speed of the fly = 200 m/s
- Total time $T = 5(\sqrt{5} - 1)$ seconds

Substituting these values: $\text{Distance}_{fly} = 200 \times 5(\sqrt{5} - 1)$ $\text{Distance}_{fly} = 1000(\sqrt{5} - 1)$ meters

This is the total distance traveled by the fly before the bus collides with the wall.

Notes

The displacement in the nth second (S_n) is given by the formula: $S_n = u + \frac{a}{2}(2n - 1)$ Where:

- u is the initial velocity.
- a is the constant acceleration.
- n is the nth second (e.g., for the first second, $n = 1$; for the third second, $n = 3$).

Example: Calculating Displacement in Specific Seconds (Starting from Rest)

Let's consider a scenario where the initial velocity (u) is 0 and the acceleration is a.

1. **Displacement in the First Second (S_1):**
   - Substitute $u = 0$ and $n = 1$ into the formula: $S_1 = 0 + \frac{a}{2}(2 \times 1 - 1)$ $S_1 = \frac{a}{2}(2 - 1)$ $S_1 = \frac{a}{2}$
2. **Displacement in the Third Second (S_3):**
   - Substitute $u = 0$ and $n = 3$ into the formula: $S_3 = 0 + \frac{a}{2}(2 \times 3 - 1)$ $S_3 = \frac{a}{2}(6 - 1)$ $S_3 = \frac{5a}{2}$
3. **Ratio of Displacement in the Third Second to the First Second ($S_3 : S_1$):**
   - The ratio is: $\frac{S_3}{S_1} = \frac{\frac{5a}{2}}{\frac{a}{2}} = 5$
   - Therefore, $S_3 : S_1 = 5 : 1$.
4. **Displacement in the Second Second (S_2):**
   - Substitute $u = 0$ and $n = 2$ into the formula: $S_2 = 0 + \frac{a}{2}(2 \times 2 - 1)$ $S_2 = \frac{a}{2}(4 - 1)$ $S_2 = \frac{3a}{2}$

Observation of Displacement Pattern

We observe the following displacements:

- $S_1 = \frac{a}{2}$
- $S_2 = \frac{3a}{2}$
- $S_3 = \frac{5a}{2}$

If we take the ratio of these displacements, we get: $S_1 : S_2 : S_3 = \frac{a}{2} : \frac{3a}{2} : \frac{5a}{2}$ Dividing by the common factor $\frac{a}{2}$, the ratio becomes: $S_1 : S_2 : S_3 = 1 : 3 : 5$ This pattern continues for subsequent seconds. For example, S_4 would be $7 \times \frac{a}{2}$, leading to a ratio of $1 : 3 : 5 : 7$, and so on.

Galileo's Rule (Law of Odd Numbers)

moves with constant acceleration, the distances covered by the body in successive equal intervals of time are in the ratio of odd numbers, i.e., 1 : 3 : 5 : 7 : ....

Conditions for Applying Galileo's Rule

For Galileo's Rule to be applicable, two conditions must be met:

1. The body must start **from rest** ($u = 0$).
2. The body must move with **constant acceleration** (a).

Significance

This rule provides a quick way to determine the ratio of distances covered in successive equal time intervals under the specified conditions. It is a direct consequence of the equations of motion and can be very useful in solving certain problems quickly. The derivation above serves as a proof of Galileo's Rule.

Problem: Distance Traveled by Accelerating Car

Question

A car uniformly accelerates from rest to 20 m/s in 10 s. What is the distance traveled?

Solution

1. **Identify Given Information:**
   - Initial velocity (U) = 0 m/s (starts from rest)
   - Final velocity (V) = 20 m/s
   - Time (T) = 10 s
2. **Calculate Acceleration (a):** We use the first equation of motion: $V = U + aT$ $20 = 0 + a(10)$ $10a = 20$ $a = \frac{20}{10} = 2$ m/s^2
3. **Calculate Distance/Displacement (S):** Now that we have acceleration, we can use the second equation of motion: $S = UT + \frac{1}{2}aT^2$ $S = (0)(10) + \frac{1}{2}(2)(10)^2$ $S = 0 + 1(100)$ $S = 100$ m

Important Distinction: Displacement vs. Distance

- The variable S in the equations of motion ($S = UT + \frac{1}{2}aT^2$) represents **displacement**, not distance.
- **When are displacement and distance equal?**
  - If an object moves in a straight line without changing its direction, the magnitude of its displacement is equal to the distance traveled.

Notes

- **When are displacement and distance different?**
  - If the car were to change direction (e.g., move forward and then come back), then the distance traveled would be greater than the magnitude of the displacement. In such cases, the motion would need to be broken into parts to calculate the total distance.

Understanding Distance vs. Displacement with Changing Direction

Problem Statement

Imagine a car (let's call it a 'Khatara Ferrari') moving at an initial speed of 50 m/s. It experiences a constant retardation (deceleration) of 5 m/s^2. The question asks to find the **distance** covered by the car in the first 20 seconds.

Initial Calculation (Displacement)

Let's first calculate the displacement using the standard kinematic equation: $S = UT + \frac{1}{2}AT^2$ Given:

- Initial velocity (U) = 50 m/s (assuming rightward as positive)
- Time (T) = 20 s
- Acceleration (A) = −5 m/s^2 (since it's retardation and acts opposite to initial velocity)

Substituting these values: $S = (50)(20) + \frac{1}{2}(-5)(20)^2$ $S = 1000 + \frac{1}{2}(-5)(400)$ $S = 1000 - (5)(200)$ $S = 1000 - 1000$ $S = 0$ m

Analysis of Displacement Result

The displacement calculated is 0 m. This might seem counterintuitive because the car was initially moving, so the distance covered cannot be zero. This result indicates that the car must have moved forward and then returned to its starting position within the 20 seconds. Displacement is a vector quantity and only considers the net change in position, while distance is a scalar quantity and accounts for the total path length traveled.

Determining the Turning Point

To find the actual distance, we need to determine if and when the car changes its direction. The car will change direction when its velocity momentarily becomes zero. We can find this time using the first kinematic equation: $V = U + AT$ Where:

- Final velocity (V) = 0 m/s (when the car stops)
- Initial velocity (U) = 50 m/s
- Acceleration (A) = −5 m/s^2

considered is 20 seconds, the car moves forward for the first 10 seconds and then moves backward for the next 10 seconds.

Calculating Distance Traveled in Each Phase

1. **Distance traveled in the first 10 seconds (forward motion):** Using the displacement formula for T = 10 s: $S_1 = UT + \frac{1}{2}AT^2$ S_1 = (50)(10) + $\frac{1}{2}$(−5)(10)^2 S_1 = 500 + $\frac{1}{2}$(−5)(100) S_1 = 500 − (5)(50) S_1 = 500 − 250 S_1 = 250 m So, the car moves 250 m forward in the first 10 seconds.

2. **Distance traveled in the next 10 seconds (backward motion):** Since the car stops at 10 seconds and then accelerates backward due to the constant negative acceleration, it will cover the same distance in the next 10 seconds, returning to its starting point. This is why the total displacement was 0 m. The distance covered backward is also 250 m.

Total Distance Calculation

The total distance is the sum of the magnitudes of the distances covered in both phases:
Total Distance = $|S_1| + |S_2|$ Total Distance = 250 m + 250 m Total Distance = 500 m

Alternatively, since the car travels 250 m forward and then 250 m backward to reach the starting point, the total distance is simply twice the distance covered in the first 10 seconds.
Total Distance = 2 × 250 m = 500 m

Conclusion

It is crucial to distinguish between distance and displacement, especially when the direction of motion changes. If a body reverses its direction during the motion, calculating displacement directly using kinematic equations might yield zero, but the distance covered will be non-zero. In such cases, it's necessary to find the time when the velocity becomes zero to determine the turning point and calculate the distance for each segment of the journey separately.

Problem: Speed of Midpoint of a Train

Problem Statement

Imagine a long train with F as the front, B as the back, and M as the midpoint. P is a fixed point on the ground.

- When the front (F) of the train crosses point P, its speed is U.
- When the back (B) of the train crosses point P, its speed is V.

Find the speed at which the midpoint (M) of the train crosses point P.

velocity, final velocity, acceleration, and displacement, as time is not given.

Step 1: Define Variables and Initial Conditions

Let:

- L be the total length of the train.
- a be the constant acceleration of the train.

When the front (F) crosses P, the speed is U. This can be considered the initial speed for the entire motion of the train past point P.

Step 2: Analyze the Motion from F crossing P to B crossing P

When the back (B) of the train crosses point P, the entire train has moved forward by its length L. The final speed at this point is V.

Using the kinematic equation $V^2 = U^2 + 2aS$:

Here:

- Initial speed = U
- Final speed = V
- Displacement (S) = L
- Acceleration = a

Substituting these values, we get: $V^2 = U^2 + 2aL$ Rearranging this equation to find $2aL$: $2aL = V^2 - U^2$ (1)

Step 3: Analyze the Motion from F crossing P to M crossing P

When the midpoint (M) of the train crosses point P, the train has moved forward by half its length, i.e., $L/2$. Let the speed of the train at this instant be V_1 (which is what we need to find).

Using the same kinematic equation $V^2 = U^2 + 2aS$:

Here:

- Initial speed = U
- Final speed = V_1
- Displacement (S) = $L/2$
- Acceleration = a

Substituting these values, we get: $V_1^2 = U^2 + 2a\left(\frac{L}{2}\right)$ $V_1^2 = U^2 + aL$ (2)

Substitute this expression for aL into equation (2): $V_1^2 = U^2 + \frac{V^2-U^2}{2}$ To simplify, find a common denominator: $V_1^2 = \frac{2U^2}{2} + \frac{V^2-U^2}{2}$ $V_1^2 = \frac{2U^2+V^2-U^2}{2}$ $V_1^2 = \frac{U^2+V^2}{2}$

Finally, take the square root to find V_1: $V_1 = \sqrt{\frac{U^2+V^2}{2}}$

Conclusion

The speed at which the midpoint of the train crosses point P is $\sqrt{\frac{U^2+V^2}{2}}$. This is a classic problem often encountered in kinematics.

Motion in Vertical Gravity

Introduction to Vertical Gravity

Motion under vertical gravity is a type of uniformly accelerated motion because the acceleration due to gravity is constant. This is similar to motion along the X-axis, but now the motion occurs along the Y-axis.

Value and Direction of Gravity

- The actual value of acceleration due to gravity (g) is approximately $9.8\ m/s^2$.
- For simplicity in calculations, we often assume $g = 10\ m/s^2$.
- Gravity always acts downwards.

Choosing a Positive Direction

When dealing with vertical motion, you have the flexibility to choose either the upward or downward direction as positive. This choice affects the sign of gravity in your calculations, but not its actual direction or magnitude.

- **If upward is chosen as positive:**
  - Gravity (g) will be considered negative (e.g., $a = -10\ m/s^2$) because it acts downwards, opposite to the chosen positive direction.
- **If downward is chosen as positive:**
  - Gravity (g) will be considered positive (e.g., $a = +10\ m/s^2$) because it acts in the same direction as the chosen positive direction.

Advice for Simplifying Calculations

- **Example 1: Ball moving upwards**
  - If a ball is thrown upwards, choose the upward direction as positive. This simplifies visualization and calculations.
- **Example 2: Ball moving downwards**
  - If a ball is falling downwards, choose the downward direction as positive. This also simplifies visualization and calculations.

This is a practical tip, not a physics rule. Physics allows any choice of positive direction, but aligning it with the primary direction of motion often leads to fewer sign errors and clearer problem-solving.

Symmetry of Upward and Downward Motion

Understanding Symmetric Motion

When an object is thrown vertically upwards, its upward and downward motions are symmetric. This means that at any given height, the speed of the object while going up is equal to its speed while coming down. The direction of velocity, however, will be opposite.

For example:

- If an object is thrown upwards with an initial speed of $20\ \text{m/s}$, it will return to the ground with a speed of $20\ \text{m/s}$. This can be understood through the principle of energy conservation, where the initial kinetic energy is converted to potential energy and then back to kinetic energy.
- If at a certain height, the upward speed is $15\ \text{m/s}$, then at the same height during its descent, the speed will also be $15\ \text{m/s}$.
- Similarly, if the upward speed at another height is $5\ \text{m/s}$, the downward speed at that same height will be $5\ \text{m/s}$.

Time of Ascent and Descent

Another aspect of this symmetry is related to time:

- The time taken for the object to reach its maximum height (time of ascent, t_{ascent}) is equal to the time taken for it to fall back to its initial position from the maximum height (time of descent, t_{descent}).
- Therefore, $t_{\text{ascent}} = t_{\text{descent}}$.
- The total time of flight (T) is the sum of the time of ascent and time of descent: $T = t_{\text{ascent}} + t_{\text{descent}}$.
- Since $t_{\text{ascent}} = t_{\text{descent}}$, each is half of the total time of flight: $t_{\text{ascent}} = t_{\text{descent}} = \frac{T}{2}$.

Notes

Time of Ascent

Understanding Time of Ascent

When a ball is thrown upwards with an initial speed 'u', it travels upwards until its velocity becomes zero at the highest point. The time taken to reach this highest point is known as the time of ascent.

To calculate the time of ascent, we can use the first equation of motion:

$v = u + at$

Here's how we apply it:

- **Initial velocity (u):** We consider the upward direction as positive. So, the initial velocity is '+u'.
- **Final velocity (v):** At the highest point, the velocity becomes zero, so '$v = 0$'.
- **Acceleration (a):** The acceleration due to gravity 'g' acts downwards. Since we've chosen upward as positive, '$a = -g$'.
- **Time (t):** This is the time of ascent we need to find.

Substituting these values into the equation:

$0 = u + (-g)t \; gt = u \; t = \frac{u}{g}$

Thus, the **time of ascent** is $\frac{u}{g}$.

Time of Descent

Understanding Time of Descent

The time of descent is the time taken for the ball to fall back to its initial position after reaching the maximum height. For motion under gravity, neglecting air resistance, the time of descent is equal to the time of ascent.

Therefore, the **time of descent** is also $\frac{u}{g}$.

Total Time of Flight

The total time of flight is the total duration for which the ball remains in the air. It is the sum of the time of ascent and the time of descent.

Total Time of Flight = Time of Ascent + Time of Descent Total Time of Flight = $\frac{u}{g} + \frac{u}{g}$
Total Time of Flight = $\frac{2u}{g}$

Notes

This section addresses the time it takes for an object, starting from rest at its maximum height, to fall through a specific height 'H'.

We use the second equation of motion:

$S = ut + \frac{1}{2}at^2$

Here's how we apply it:

- **Initial moment:** The ball is at its highest point, where its velocity is zero.
- **Final moment:** The ball has fallen through a height 'H'.
- **Direction convention:** We choose the downward direction as positive for this part of the motion.
- **Displacement (S):** The ball falls downwards by 'H', so '$S = +H$'.
- **Initial velocity (u):** At the highest point, the initial velocity for this downward motion is '0'.
- **Acceleration (a):** The acceleration due to gravity 'g' acts downwards. Since we've chosen downward as positive, '$a = +g$'.
- **Time (t):** This is the time we need to find.

Substituting these values into the equation:

$H = (0)t + \frac{1}{2}(g)t^2 \; H = \frac{1}{2}gt^2$

This is a crucial result: if an object starts from rest and falls through a height 'H', the time 't' taken is given by:

$t = \sqrt{\frac{2H}{g}}$

This result is fundamental for understanding vertical motion under gravity when starting from rest.

Maximum Height Reached

Determining the Maximum Height Reached

There are multiple methods to calculate the maximum height ('H_{max}') reached by a ball thrown upwards with initial speed 'u'.

Method 1: Using Time of Ascent in Displacement Equation

We know that the time of ascent is $t = \frac{u}{g}$. We can use the equation for displacement when starting from rest and falling through a height:

the starting point, and the time taken for this fall is the time of descent, which is equal to the time of ascent, $\frac{u}{g}$.

Substituting $t = \frac{u}{g}$ into the equation:

$$H_{max} = \frac{1}{2}g\left(\frac{u}{g}\right)^2 \; H_{max} = \frac{1}{2}g\left(\frac{u^2}{g^2}\right) \; H_{max} = \frac{u^2}{2g}$$

Method 2: Using the Third Equation of Motion for Ascent

Consider the upward motion (ascent) from the initial point to the maximum height.

We use the third equation of motion:

$$v^2 = u^2 + 2aS$$

Here's how we apply it:

- **Initial velocity (u):** The ball is thrown upwards with speed 'u'. We choose upward as positive, so '$u = +u$'.
- **Final velocity (v):** At the maximum height, the velocity becomes zero, so '$v = 0$'.
- **Acceleration (a):** The acceleration due to gravity 'g' acts downwards. Since we've chosen upward as positive, '$a = -g$'.
- **Displacement (S):** This is the maximum height 'H_{max}' we need to find.

Substituting these values into the equation:

$$0^2 = u^2 + 2(-g)H_{max} \; 0 = u^2 - 2gH_{max} \; 2gH_{max} = u^2 \; H_{max} = \frac{u^2}{2g}$$

Both methods yield the same result. This is another important result to remember: if a ball is thrown upwards with speed 'u', it will reach a maximum height of $\frac{u^2}{2g}$.

Problem: Particle Dropped from Height

Problem Statement

A particle is dropped from the top of a tall building (e.g., Burj Khalifa). There is a window of length 50 meters. The top of the window is located at a height H from the top of the building. The particle crosses the entire window in 2 seconds. We need to find the value of H.

Given Information

- Length of the window (PQ) = 50 meters
- Time taken to cross the window (T_{PQ}) = 2 seconds

Notes

Imagine a particle starting from point A (top of the building). It reaches point P (top of the window) after some time T. The distance covered is H. Then, it travels from P to Q (bottom of the window), covering a distance of 50 meters, in 2 seconds. The total distance from A to Q is H + 50, and the total time taken is T + 2 seconds.

Key Concept: Equation of Motion for Free Fall

For an object dropped from rest, the distance fallen (h) in time (t) under constant acceleration due to gravity (g) is given by: $h = \frac{1}{2}gt^2$ This is a crucial result for solving problems involving free fall.

Step-by-Step Solution

Applying the Equation of Motion

1. **From A to P:** The particle falls a distance H in time T. Using the equation $h = \frac{1}{2}gt^2$: $H = \frac{1}{2}gT^2$ $(Equation\ 1)$
2. **From A to Q:** The particle falls a total distance of H + 50 in a total time of T + 2 seconds. Using the equation $h = \frac{1}{2}gt^2$: $H + 50 = \frac{1}{2}g(T + 2)^2$ $(Equation\ 2)$

Solving the System of Equations

Substitute Equation 1 into Equation 2: $\frac{1}{2}gT^2 + 50 = \frac{1}{2}g(T + 2)^2$ Expand the term $(T + 2)^2$: $\frac{1}{2}gT^2 + 50 = \frac{1}{2}g(T^2 + 4T + 4)$ Distribute $\frac{1}{2}g$ on the right side: $\frac{1}{2}gT^2 + 50 = \frac{1}{2}gT^2 + \frac{1}{2}g(4T) + \frac{1}{2}g(4)$ $\frac{1}{2}gT^2 + 50 = \frac{1}{2}gT^2 + 2gT + 2g$ Subtract $\frac{1}{2}gT^2$ from both sides: $50 = 2gT + 2g$ Factor out $2g$: $50 = 2g(T + 1)$ Now, substitute the value of g = 10 m/s^2: $50 = 2(10)(T + 1)$ $50 = 20(T + 1)$ Divide by 20: $\frac{50}{20} = T + 1$ $2.5 = T + 1$ Solve for T: $T = 2.5 - 1$ $T = 1.5$ seconds

Calculating H

Now that we have the value of T, we can find H using Equation 1: $H = \frac{1}{2}gT^2$ Substitute g = 10 m/s^2 and T = 1.5 seconds: $H = \frac{1}{2}(10)(1.5)^2$ $H = 5(2.25)$ $H = 11.25$ meters

Final Answer

The value of H is 11.25 meters.

Problem: Ball Dropped from Tower

Problem Statement

A ball is dropped from the top of a tower of height H. It takes a total time T to reach the ground. The problem states that the ball travels half the total distance ($H/2$) in the *last second* of its fall.

- Point A: Top of the tower
- Point B: Bottom of the tower (ground)
- Point M: Midpoint of the tower's height, such that AM = MB = $H/2$.

The problem implies that the ball falls from M to B (the last half of the distance) in the last 1 second. This means:

- Time taken to fall from A to B = T seconds.
- Time taken to fall from A to M = $T - 1$ seconds.
- Distance fallen from A to B = H.
- Distance fallen from A to M = $H/2$.

Equations of Motion

We use the equation of motion for an object under free fall, starting from rest ($u = 0$): $s = ut + \frac{1}{2}gt^2$ Since $u = 0$, this simplifies to: $s = \frac{1}{2}gt^2$

1. **For the entire fall (A to B):** The ball falls a height H in time T. $H = \frac{1}{2}gT^2$ (Equation 1)
2. **For the first half of the distance (A to M):** The ball falls a height $H/2$ in time $T - 1$. $\frac{H}{2} = \frac{1}{2}g(T-1)^2$ (Equation 2)

Solving the Equations

To find the total time T, we can divide Equation 1 by Equation 2:

$$\frac{H}{\frac{H}{2}} = \frac{\frac{1}{2}gT^2}{\frac{1}{2}g(T-1)^2}$$

This simplifies to:

$$2 = \frac{T^2}{(T-1)^2}$$

Now, take the square root of both sides:

$$\sqrt{2} = \frac{T}{T-1}$$

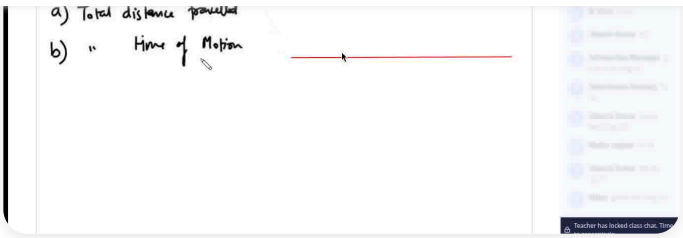
Rearrange the equation to solve for T:

$$\sqrt{2}(T-1) = T\ \sqrt{2}T - \sqrt{2} = T\ \sqrt{2}T - T = \sqrt{2}\ T(\sqrt{2}-1) = \sqrt{2}\ T = \frac{\sqrt{2}}{\sqrt{2}-1}$$

This is the total time T taken for the ball to fall from the top of the tower to the ground.

Introduction to Bouncing Ball Problem

Notes



Problem Statement

A ball is dropped from a height H_0. After every collision with the ground, the speed of the ball becomes half. We need to find the total distance traveled and the total time of motion.

Initial Conditions and Observations

- The ball is dropped from an initial height H_0.
- Upon striking the ground for the first time, let its speed be V_0.
- After the first collision, it rebounds with a speed $V_1 = V_0/2$.
- It then rises to a certain height and falls back, striking the ground again with speed V_1.
- After the second collision, it rebounds with a speed $V_2 = V_1/2 = V_0/4$.
- This pattern continues indefinitely, with the speed becoming half after each collision: $V_n = V_0/2^n$.

Key Physics Relations

We use the following relations for motion under gravity:

- If an object is dropped from height H, its speed upon landing is $V = \sqrt{2gH}$. Conversely, if an object lands with speed V, the height it fell from is $H = V^2/(2g)$.
- If an object is thrown upwards with speed U, it reaches a maximum height $H = U^2/(2g)$ and returns to the ground with the same speed U.
- The time of flight for an object thrown upwards with speed U and returning to the same level is $2U/g$. The time to reach the peak (or fall from the peak) is U/g.

Initial Fall

For the first fall from height H_0, the ball strikes the ground with speed V_0. Using the relation $H = V^2/(2g)$: $H_0 = \frac{V_0^2}{2g}$ This can also be derived from energy conservation: $mgH_0 = \frac{1}{2}mV_0^2$, which simplifies to $H_0 = \frac{V_0^2}{2g}$.

Calculating Total Distance Traveled

Distance Calculation Strategy

distance of $2h_1$.

- After the second bounce, it goes up to h_2 and comes down, covering $2h_2$, and so on.
- The total distance will be $D_{total} = H_0 + 2h_1 + 2h_2 + 2h_3 + \ldots$

Heights after Bounces

Let V_n be the speed with which the ball rebounds after the n-th collision. The height it reaches, h_n, is given by $h_n = V_n^2/(2g)$.

- **First rebound:** The ball rebounds with speed $V_1 = V_0/2$. The height reached is $h_1 = V_1^2/(2g)$. The distance covered in this phase (up and down) is $2h_1 = 2\left(\frac{V_1^2}{2g}\right) = \frac{V_1^2}{g}$.
- **Second rebound:** The ball rebounds with speed $V_2 = V_1/2 = V_0/4$. The height reached is $h_2 = V_2^2/(2g)$. The distance covered is $2h_2 = 2\left(\frac{V_2^2}{2g}\right) = \frac{V_2^2}{g}$.
- **Third rebound:** The ball rebounds with speed $V_3 = V_2/2 = V_0/8$. The height reached is $h_3 = V_3^2/(2g)$. The distance covered is $2h_3 = 2\left(\frac{V_3^2}{2g}\right) = \frac{V_3^2}{g}$.

Summing the Distances

The total distance traveled is: $D_{total} = H_0 + \frac{V_1^2}{g} + \frac{V_2^2}{g} + \frac{V_3^2}{g} + \ldots$ Substitute the values of V_n in terms of V_0:

- $V_1 = V_0/2 \implies V_1^2 = V_0^2/4$
- $V_2 = V_0/4 \implies V_2^2 = V_0^2/16$
- $V_3 = V_0/8 \implies V_3^2 = V_0^2/64$

So, the expression becomes: $D_{total} = H_0 + \frac{V_0^2}{4g} + \frac{V_0^2}{16g} + \frac{V_0^2}{64g} + \ldots$ We know $H_0 = V_0^2/(2g)$. Substitute this into the equation: $D_{total} = \frac{V_0^2}{2g} + \frac{V_0^2}{4g} + \frac{V_0^2}{16g} + \frac{V_0^2}{64g} + \ldots$

Recognizing a Geometric Progression

The terms from the second term onwards form an infinite geometric progression (GP): $S = \frac{V_0^2}{4g} + \frac{V_0^2}{16g} + \frac{V_0^2}{64g} + \ldots$ Here, the first term $a = \frac{V_0^2}{4g}$ and the common ratio $r = \frac{1}{4}$. The sum of an infinite GP is given by $S = \frac{a}{1-r}$. $S = \frac{\frac{V_0^2}{4g}}{1-\frac{1}{4}} = \frac{\frac{V_0^2}{4g}}{\frac{3}{4}} = \frac{V_0^2}{4g} \times \frac{4}{3} = \frac{V_0^2}{3g}$

Total Distance

Now, add the initial fall distance H_0 to the sum S: $D_{total} = H_0 + S = \frac{V_0^2}{2g} + \frac{V_0^2}{3g}$ To combine these, find a common denominator: $D_{total} = \frac{3V_0^2}{6g} + \frac{2V_0^2}{6g} = \frac{5V_0^2}{6g}$ Since $H_0 = V_0^2/(2g)$, we can write $V_0^2 =$

Notes

Calculating Total Time of Motion

Time Calculation Strategy

- The initial fall takes time T_0.
- After the first bounce, the ball goes up and comes down, taking time $2T_1$.
- After the second bounce, it goes up and comes down, taking time $2T_2$, and so on.
- The total time will be $T_{total} = T_0 + 2T_1 + 2T_2 + 2T_3 + \ldots$

Time for Each Phase

- **Initial fall:** The ball falls from H_0 and lands with speed V_0. The time taken T_0 can be found using $V_0 = gt_0$, so $T_0 = V_0/g$. Alternatively, $H_0 = \frac{1}{2}gT_0^2$, so $T_0 = \sqrt{\frac{2H_0}{g}}$.
- **Time for ascent/descent after bounces:** If a ball is thrown up with speed U, the time to reach the peak (or fall from the peak) is U/g. So, the total time for one bounce cycle (up and down) is $2U/g$.

Let V_n be the speed with which the ball rebounds after the n-th collision.

- **First rebound:** Rebounds with speed $V_1 = V_0/2$. The time for this cycle (up and down) is $2T_1 = 2\left(\frac{V_1}{g}\right) = \frac{2V_1}{g}$.
- **Second rebound:** Rebounds with speed $V_2 = V_0/4$. The time for this cycle is $2T_2 = 2\left(\frac{V_2}{g}\right) = \frac{2V_2}{g}$.
- **Third rebound:** Rebounds with speed $V_3 = V_0/8$. The time for this cycle is $2T_3 = 2\left(\frac{V_3}{g}\right) = \frac{2V_3}{g}$.

Summing the Times

The total time of motion is: $T_{total} = T_0 + \frac{2V_1}{g} + \frac{2V_2}{g} + \frac{2V_3}{g} + \ldots$ Substitute the values of V_n in terms of V_0:

- $V_1 = V_0/2$
- $V_2 = V_0/4$
- $V_3 = V_0/8$

So, the expression becomes: $T_{total} = \frac{V_0}{g} + \frac{2(V_0/2)}{g} + \frac{2(V_0/4)}{g} + \frac{2(V_0/8)}{g} + \ldots$ $T_{total} = \frac{V_0}{g} + \frac{V_0}{g} + \frac{V_0}{2g} + \frac{V_0}{4g} + \ldots$

Recognizing a Geometric Progression

given by $S' = \frac{a}{1-r}$. $S' = \frac{g}{1-\frac{1}{2}} - \frac{g}{\frac{1}{2}} - \frac{-1.0}{g}$

Total Time

Now, add the initial fall time T_0 to the sum S': $T_{total} = T_0 + S' = \frac{V_0}{g} + \frac{2V_0}{g} = \frac{3V_0}{g}$ We know $V_0 = \sqrt{2gH_0}$. Substitute this back: $T_{total} = \frac{3\sqrt{2gH_0}}{g} = 3\sqrt{\frac{2H_0}{g}}$ So, the total time of motion is $3\sqrt{\frac{2H_0}{g}}$.

Clarification on Previous Problem (H + 50)

Understanding the H + 50 Equation

During a previous problem, a student asked for clarification on the equation involving "H + 50". This equation represents the total height through which an object has fallen. If the initial height is 'H' and an additional 50 units of distance are covered, the total height fallen is H + 50.

Time Calculation

If the initial time is 'T' and an additional 2 seconds pass, the total time elapsed becomes T + 2 seconds.

Applying Kinematic Equations

The total height fallen (H + 50) is related to the total time (T + 2) by the kinematic equation for free fall: H + 50 = $\frac{1}{2}g(T + 2)^2$ Where:

- H + 50 is the total height fallen.
- g is the acceleration due to gravity.
- T + 2 is the total time elapsed.

Expansion and Substitution

To solve for unknown variables, this equation is expanded. If we know that $H = \frac{1}{2}gT^2$ (from an earlier part of the problem), we can substitute this into the expanded equation: $\frac{1}{2}gT^2$ + 50 = $\frac{1}{2}g(T^2 + 4T + 4)$ This equation can then be solved for 'T' or other variables through algebraic manipulation.

Understanding Retardation vs. Negative Acceleration

Definition of Retardation

Retardation specifically means **slowing down**. It occurs when the speed of an object decreases. For example, if a ball is thrown upwards, its speed decreases as it moves against gravity, so it experiences retardation.

always retardation. This is **incorrect**.

Dependence on Sign Convention

Whether acceleration is positive or negative depends entirely on the chosen sign convention for direction.

Consider a particle released from the top:

- **If upward is chosen as positive:** The acceleration due to gravity will be $-g$ (negative).
- **If downward is chosen as positive:** The acceleration due to gravity will be $+g$ (positive).

In both cases, the particle's speed will **increase** as it falls. Therefore, even when acceleration is negative (e.g., $-g$ when upward is positive), the object is speeding up, not slowing down. This means it is **not** experiencing retardation.

Key Distinction

- **Retardation:** Speed is decreasing.
- **Negative Acceleration:** Acceleration is in the direction opposite to the chosen positive direction. This does not necessarily mean the object is slowing down; it could still be speeding up if its velocity is also in the negative direction.

Homework and Revision Strategy

Daily Practice Problems (DPPs)

Students will receive two Daily Practice Problems (DPPs) today. It is crucial to solve these to avoid a backlog. Each DPP will contain approximately 25 questions.

Importance of Solving Homework

Solving homework is essential for reinforcing concepts and improving problem-solving skills. The questions provided are specifically designed for students preparing for competitive exams like JEE, considering their existing knowledge base.

Revision Strategy

1. **Redo Each Question:** Go through all the questions discussed in the session and solve them again independently.
2. **Revise Concepts:** Review the session content and notes to ensure a clear understanding of all concepts.
3. **Solve Homework:** After revising and redoing the session problems, attempt the DPPs.

Notes